Automotive Cyber Security Academy 2023

Basic Information Technology Security

Topic: Motivation for Security

Automotive Cyber Security Academy

1 2 9 0 FACULDADE DE CIÊNCIAS E TECNOLOGIA UNIVERSIDADE D COIMBRA

dei departamento de engenharia informática

30/04/23

Outline

- What is Cyber Security
- Why we need Cyber Security
- Some exemples of attacks
- My Motivation to work on this topic

What is Cyber Security ?

- The cyber space includes the Internet, the data that is distributed to be shared with other people
- The devices that connect to the cyber space have heterogeneous technologies (smartwatches, mobile phones)
- So, Cyber Security is:
  - **Set of <u>technologies, processes and procedures</u> to protect the cyber space from potential threats and attacks**
  - **It aims to protect networks, computers, other devices, programs, data and the interaction of people with the digital world**

Why we need Cyber Security ?

The cybersecurity workforce shortage and skills gap is a major concern for both economic development and national security. By looking into the problem, ENISA identified Europe's need for a comprehensive approach to define a set of cybersecurity roles and skills that could be leveraged to reduce the shortage and the skills gap. ENISA has worked on the development of

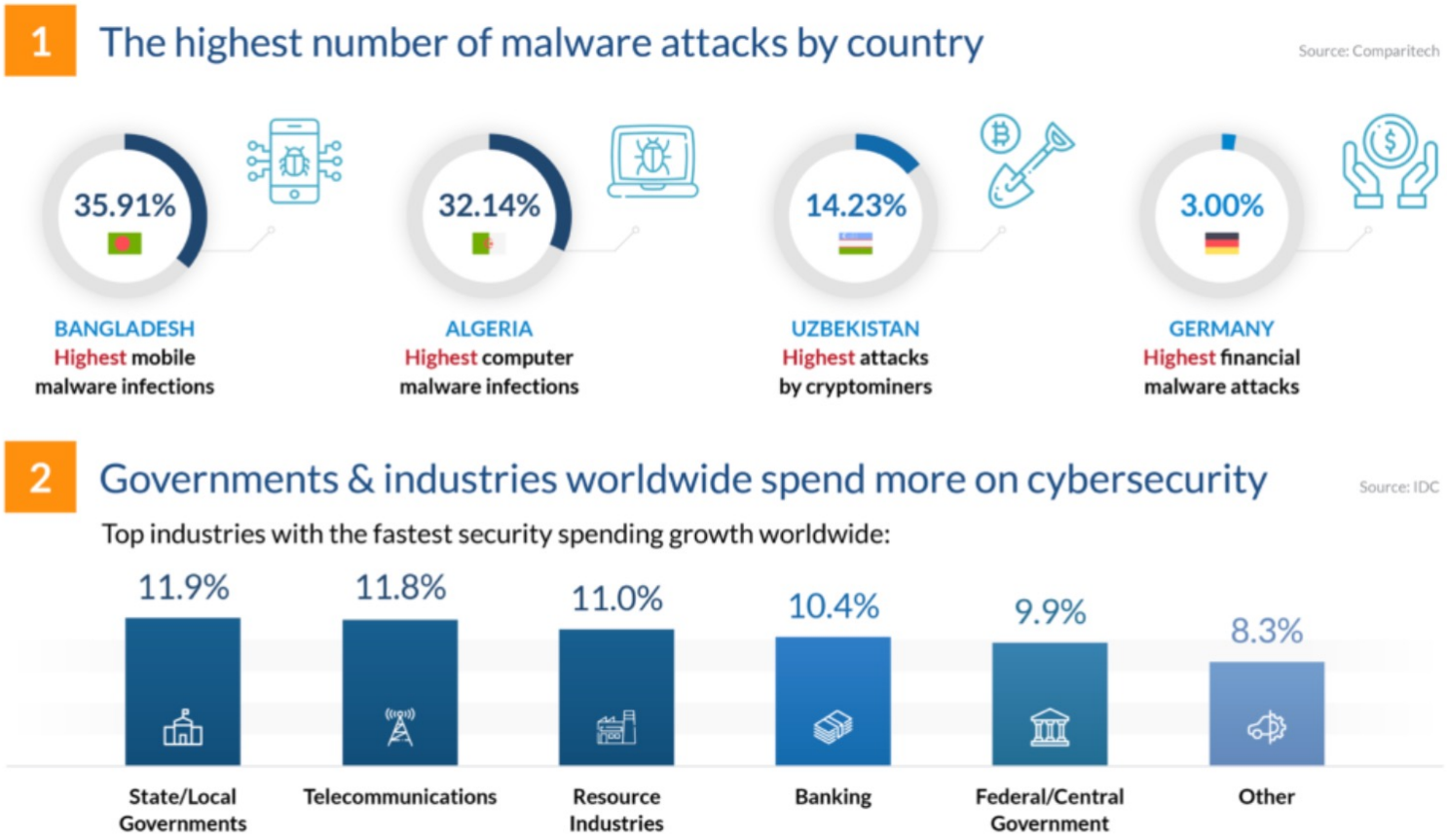
ENISA European Cybersecurity Skills Framework

- There is a lack of cyber security experts and not only at European level

State of Cybersecurity 2022:
Cyber Workforce Challenges

ISACA

Why we need Cyber Security ?



Why we need Cyber Security ?

Image Credit: Midjourney

Source: VentureBeat

Examples of Attacks
Exposure to Security Vulnerabilities

Oliveira do Hospital

Municípios vulneráveis ao cibercrime

DR

Tal como o DIÁRIO AS BEIRAS revelou, o Município de Oliveira do Hospital foi, recentemente, alvo de ciberataque

●●● Pelo menos duas dezenas de autarquias portuguesas registaram, nos últimos quatro anos, ciberataques aos seus sistemas informáticos, mas a organização da sociedade CpC: Cidadãos pela Cibersegurança regista que o investimento que fazem em cibersegurança é quase nulo. Oliveira do Hospital (há cerca de uma semana), foi o município que se juntou a um lote de outros da região – como Figueira da Foz, Mira, Pampilhosa da Serra – alvos de ciberataques desde 2019, mas também Faro, Loures, Barreiro e Barcelos.

Embora Portugal tenha uma estratégia nacional de cibersegurança desde 2015 – em processo de atualização sob a designação de "Segurança do Ciberespaço 2019-2023" – "após consulta dos orçamentos de algumas das maiores câmaras municipais, constata-se a ausência de uma rubrica dedicada à cibersegurança", denunciou ontem, em comunicado, a CpC.

Número de queixas

Por outro lado, consultando a base de contratação pública, deteta-se que só nove autarquias portuguesas investiram em cibersegurança, em cumprimento do decreto lei nº 65/2021 de 30 de julho. A única da região é a Figueira da Foz, para "implantação do Regime Jurídico de Cibersegurança", em que investiu 17.650 euros há cerca de um ano.

O Gabinete de Cibercrime da Procuradoria Geral da República (PGR) publicou em fevereiro último uma síntese das denúncias de cibercrime em 2022 referindo que "foram recebidas 2.124 denúncias, quando em 2021 tinham sido recebidas 1.160". Quanto a denúncias registou-se um aumento de 73,58% de 2021 para 2022. Em termos médios – com oscilações anuais, desde 2019 – são recebidas o dobro das denúncias do ano anterior". Foram encaminhadas para abertura de inquérito ou para a Polícia Judiciária 359.

Às instituições resta o apoio que é prestado pelo Centro Nacional de Cibersegurança Português (CNCS), designadamente no capítulo da formação mas, de acordo com a CpC, só três autarquias em 308 o fazem e apenas uma (Almada) fez teste de intrusão.

| António Rosado

Portuguese Municipalities exposed to Cybercrime
Low investment from public bodies in a cyber security culture.

12/April/2023 (source)

Examples of Attacks
Exposure to Security Vulnerabilities

Home > Data breach incident management and recovery

BRENT HOFACKER - FOTOLIA

NEWS

KFC, Pizza Hut data stolen in January ransomware attack

Yum!, the parent organisation behind KFC and Pizza Hut in the UK, has disclosed that employee data was accessed and exfiltrated in a January 2023 ransomware attack

By **Alex Scroxton,** Security Editor

Published: **11 Apr 2023 14:45**

Data Exfiltration

Do you still want to order ?

11/April/2023 (source)

Examples of Attacks
Exposure to Security Vulnerabilities

Home > Data breach incident management and recovery

JAMES STEIDL - FOTOLIA

NEWS

3CX incident may be world's first double supply chain attack

It's supply chain attacks all the way down as Mandiant publishes information suggesting that the 3CX software supply chain compromise was initiated via a prior software supply chain compromise

By **Alex Scroxton,** Security Editor

Published: **20 Apr 2023 14:45**

Supply Chain Attacks

With the installation of Malware

20/April/2023 (source)

Examples of Attacks
Exposure to Security Vulnerabilities

Home > News > Games

Among Us DDoS Attack Finally Neutralized

It took three days, but the servers are back online and stable again.

By Matthew Humphries Updated March 29, 2022

UPDATE 3/29: Mitigating the DDoS attack took longer than expected, but the Among Us Twitter account finally gave players the all clear to start playing the game again last night, along with the hope that "the sabotages don't come back."

DDoS Attacks to game servers

29/March/2022 (source)

Examples of Attacks
Exposure to Security Vulnerabilities

- We need to stop here!
- This is an endless list…

My Motivation…

- I believe in the power of education and knowledge sharing
- I believe people with the (correct) information can correct mistakes and reduce the probability of security threats
- I believe (good) cooperation can contribute to the development of security solutions for future networks.

My Motivation…

ENISA European Cybersecurity Skills Framework Roles Profile

Reflection work

- What’s your motivation to learn security ?
- Do you want to attack ?
- Do you want to defend ?
- How can you contribute to enhance security?

Thanks

Automotive Cyber Security Academy

The NEW and IMPROVED - ACSA 2023 is here!